

City General Hospital & Research Centre

Department of Clinical Pathology & Diagnostic Services

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NABL Accredited Lab | Lab ID: CGH-LAB-2024

COMPREHENSIVE MEDICAL LABORATORY REPORT

Patient Name:	Rajesh Kumar Nair	Report No:	CGH-2024-08741
Age / Sex:	52 Years / Male	Sample ID:	SMP-20241112-003
DOB:	14-Mar-1972	Collected:	12-Nov-2024, 07:45 AM
Patient ID:	PID-00938472	Reported:	12-Nov-2024, 02:30 PM
Referring Doctor:	Dr. Priya Menon, MD (Internal Medicine)	Ward / OPD:	OPD - Cardiology
Contact:	+91-98470-12345	Insurance ID:	PMJAY-2024-KL-88821

CLINICAL HISTORY & DIAGNOSIS

Known case of Type 2 Diabetes Mellitus (T2DM) and Hypertension on regular medications. Presenting with complaints of fatigue, polyuria, and mild pedal edema for the past 3 weeks. Referred for routine diabetic workup, kidney function assessment, and lipid profile.

CURRENT MEDICATIONS

Medication	Dose	Frequency	Route
Metformin 500mg	500 mg	Twice daily	Oral
Glimepiride	2 mg	Once daily (morning)	Oral
Amlodipine	5 mg	Once daily	Oral
Telmisartan	40 mg	Once daily (evening)	Oral
Atorvastatin	20 mg	Once daily at bedtime	Oral
Aspirin (Ecosprin)	75 mg	Once daily	Oral

LABORATORY INVESTIGATION RESULTS

A. BLOOD GLUCOSE PROFILE

Test	Result	Unit	Reference Range	Status
Fasting Blood Glucose (FBG)	214	mg/dL	70 - 100	HIGH *
Post-Prandial Glucose (PPBG)	318	mg/dL	< 140	HIGH **
HbA1c (Glycated Haemoglobin)	9.2	%	< 5.7 (Normal) / < 7.0 (DM target)	HIGH *

B. KIDNEY FUNCTION TEST (KFT)

Test	Result	Unit	Reference Range	Status
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Blood Urea Nitrogen (BUN)	28	mg/dL	7 - 25	HIGH *
Serum Creatinine	1.6	mg/dL	0.7 - 1.3	HIGH *
eGFR (CKD-EPI)	48	mL/min/1.73m2	> 60	LOW **
Uric Acid	6.8	mg/dL	3.4 - 7.0	Normal
Sodium (Na+)	138	mEq/L	136 - 146	Normal
Potassium (K+)	5.1	mEq/L	3.5 - 5.1	Normal

C. LIPID PROFILE

Test	Result	Unit	Reference Range	Status
Total Cholesterol	242	mg/dL	< 200	HIGH *
LDL Cholesterol	158	mg/dL	< 100 (DM patients)	HIGH **
HDL Cholesterol	38	mg/dL	> 40	LOW *
Triglycerides	286	mg/dL	< 150	HIGH **
VLDL Cholesterol	57	mg/dL	< 30	HIGH *

D. COMPLETE BLOOD COUNT (CBC)

Test	Result	Unit	Reference Range	Status
Haemoglobin (Hb)	11.2	g/dL	13.5 - 17.5	LOW *
WBC Count	8,400	cells/uL	4,500 - 11,000	Normal
Platelet Count	1,98,000	cells/uL	1,50,000 - 4,00,000	Normal
RBC Count	4.0	million/uL	4.5 - 5.5	LOW *
MCV	76	fL	80 - 100	LOW

CLINICAL IMPRESSIONS & RECOMMENDATIONS

- Poorly Controlled Type 2 Diabetes Mellitus:** HbA1c of 9.2% indicates significantly poor glycaemic control. Urgent medication review and dietary counselling recommended. Consider insulin initiation or regimen intensification.
- Early Diabetic Nephropathy (CKD Stage 3a):** eGFR of 48 mL/min/1.73m2 with elevated creatinine strongly suggests early chronic kidney disease. Nephrologist referral advised. Monitor urine albumin-creatinine ratio.
- Dyslipidaemia:** Significantly elevated LDL, triglycerides, and VLDL with low HDL in a diabetic patient confers high cardiovascular risk. Consider upscaling statin therapy to high-intensity (Rosuvastatin 40mg). Fenofibrate addition may be warranted for elevated TG.
- Microcytic Anaemia:** Low Hb with low MCV and RBC count suggestive of iron deficiency anaemia. Serum iron, TIBC, and ferritin levels advised. Dietary iron supplementation recommended.
- URGENT FLAG: Combination of poor glycaemic control, early nephropathy, and high cardiovascular risk profile warrants immediate specialist review. Dose adjustment of Metformin required given eGFR < 60.**

SUGGESTED FOLLOW-UP

Investigation	Suggested Timeline
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Urine Albumin-Creatinine Ratio (uACR)	Within 1 week
Serum Iron, TIBC, Ferritin	Within 1 week
Repeat KFT	4 weeks
Repeat HbA1c	3 months
Ophthalmology review (Diabetic Retinopathy screening)	Within 1 month
Cardiology review & ECG	Within 2 weeks

Verified & Reported by:

Dr. Ananya Krishnamurthy

MD Pathology | Reg. No: KMC-2018-4421

City General Hospital, Thiruvananthapuram

Lab Supervisor:

Mrs. Sreelakshmi Nair, MLT

DMLT, B.Sc MLT | Staff ID: CGH-LAB-052

City General Hospital, Thiruvananthapuram

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